Durham University

# Lecture 8: Strong **NP**-completeness, **NP** and **coNP**

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# KNAPSACK

KNAPSACK
**Instance**: A set of items $I = \{i_1, \ldots i_n\}$ where each item $i_j$ has value $v_j$ and weight $w_j$ (values and weights are positive integers); and a target value $t$ and a weight upper limit $L$.
**Question:** Is there a subset $T$ of $I$ with total value at least $t$ and total weight at most $L$.

## Theorem

KNAPSACK *is* **NP***-complete.*

- Step 1. The problem is in **NP**: a subset $T$ is the certificate.
- Step 2. To show completeness, reduce SUBSET SUM to KNAPSACK. We will use the method of "restriction".

# Reduction

SUBSET SUM
**Instance**: A set of positive integers $S = \{n_1, n_2, \ldots, n_k\}$, and a target integer $t'$.
**Question:** Is there a subset $T \subseteq S$, such that $\sum_{i \in T} n_i = t'$?

- Consider instances of KNAPSACK such that $v_j = w_j (= n_j)$ for all $j$, and $t = L (= t')$. Then we get exactly SUBSET SUM.
- Indeed, we have both $\sum_{j \in T} v_j \geq t$ and $\sum_{j \in T} w_j \leq L$ if and only if $\sum_{j \in T} n_j = t'$.

# A polynomial algorithm?

KNAPSACK can be solved in $O(nL)$ time using dynamic programming

- Create a $(L+1) \times (n+1)$ matrix $M$
- Set $M(w, 0) = M(0, i)$ to 0 for all $w \in \{1, 2, \ldots, L\}$ and all $i \in \{1, 2, \ldots, n\}$
- For $i = 0, 1, \ldots, n-1$, set entry $M(w, i+1)$ as follows

$$M(w, i+1) = \max\{M(w, i),\ M(w - w_{i+1}, i) + v_{i+1}\}$$

($M(w, i)$ is the largest total value obtainable by selecting from the first $i$ items with weight limit $w$)

- Answer yes if any entry $\geq t$, otherwise no

# Example construction

$M(w, i+1) = \max\{M(w, i),\ M(w - w_{i+1}, i) + v_{i+1}\}$

Let $n = 4$; values $1, 3, 4, 2$; weights $1, 1, 3, 2$; target $t = 7$ and limit $L = 5$.

| weight limit $w$ | max total value from first $i$ items | | | | |
|---|---|---|---|---|---|
| | $i = 0$ | $i = 1$ | $i = 2$ | $i = 3$ | $i = 4$ |
| 0 | | | | | |
| 1 | | | | | |
| 2 | | | | | |
| 3 | | | | | |
| 4 | | | | | |
| 5 | | | | | |

# Pseudo-polynomial time

This algorithm is not sufficient to show that KNAPSACK is in **P** ! Why?

The length of the input to KNAPSACK is in $O(n \log L) \Rightarrow nL$ is not bounded by a polynomial function of the input length.
(This is a typical source of confusion in amateur "proofs" that **P** = **NP**: numbers must be encoded in binary, not in unary!)

However, if we restrict KNAPSACK to instances such that $L \leq p(n)$ for some polynomial $p$, then we obtain a problem in **P**.

- An algorithm which is polynomial in the size of the numbers in the input is called pseudo-polynomial
- NP-complete problems which remain NP-complete when all numbers are bounded by some polynomial in the length of the input are called strongly NP-complete
  (otherwise they are called weakly NP-complete)

# Strongly **NP**-complete problems

- Any **NP**-complete problem without numerical data is strongly **NP**-complete: e.g., SATISFIABILITY, HAMILTONIAN CYCLE, 3-COLOURABILITY, VERTEX COVER, INDEPENDENT SET, CLIQUE...
- TSP (D) is strongly **NP**-complete, since it is **NP**-complete even if all distances are 1 and 2
  (see reduction from HAMILTONIAN CYCLE).
- SUBSET SUM is not strongly **NP**-complete, it is a subproblem of KNAPSACK (i.e. "easier" than KNAPSACK).

# Examples

PARTITION
**Instance**: A multi-set $S$ of positive integers (i.e. some integers may appear multiple times in $S$).
**Question:** Is there a partition of $S$ into two subsets $S_1, S_2$ such that $\sum_{s \in S_1} s = \sum_{s \in S_2} s$?

- This problem is (weakly) **NP**-complete
- A dynamic programming algorithm with running time $O(|S| \cdot \sum_{s \in S} s)$
- The length of the input is $O(|S| + \sum_{s \in S} \log s)$
- This running time is pseudo-polynomial (but not polynomial!)
  - if we encode the input numbers in unary (large encoding) $\Rightarrow$ running time is small (compared with the input size)
  - if we encode the input numbers in binary (succinct encoding) $\Rightarrow$ running time is large (compared with the input size)

# Examples

3-PARTITION
**Instance**: A multi-set $S$ of with $n = 3m$ positive integers.
**Question:** Is there a partition of $S$ into $m$ subsets $S_1, S_2, \ldots, S_m$, such that $\sum_{s \in S_i} s = \sum_{s \in S_j} s$ for every $i, j \leq m$?

- Similar problem to PARTITION, but still very different !
- It is strongly **NP**-complete
  - the order of the time complexity does not change much if we represent the numbers in unary (large encoding) or in binary (succinct encoding)
- Let $B = \frac{1}{m} \cdot \sum_{s \in S} s$
- It remains (strongly) **NP**-complete even if $\frac{B}{4} < s < \frac{B}{2}, \forall s \in S$
  - in this case every set $S_i$ has exactly 3 elements
- There exists no (pseudo-) polynomial algorithm (unless **P**= **NP**)

# **NP** and **coNP**

For a language (i.e. problem) $\mathcal{L}$ over alphabet $\Sigma$, let $\overline{\mathcal{L}}$ denote the *complement* $\Sigma^* \setminus \mathcal{L}$ of $\mathcal{L}$ (that is, $x \in \overline{\mathcal{L}} \Leftrightarrow x \notin \mathcal{L}$).

## Definition

The class of languages $\mathcal{L}$ such that $\overline{\mathcal{L}}$ has a polynomial-time verifier is called **coNP**.

In other words, a problem belongs to **coNP** if the **no**-instances have succinct certificates

Example:

NO HAMILTONIAN CYCLE
**Instance**: A graph $G$
**Question:** Is it true that $G$ has no Hamiltonian cycle?

- A Hamiltonian Cycle (if it exists!) can be verified efficiently

$\Rightarrow$ a Hamiltonian Cycle is a certificate for no-instances

# Example: TAUTOLOGY

TAUTOLOGY
**Instance**: A DNF-formula $f$
**Question:** Is it true that $f$ is a tautology, that is, is $f$ satisfied by all assignments?

- A truth assignment that makes $f$ FALSE can be verified efficiently

$\Rightarrow$ such a truth assignment is a certificate for no-instances

- Is TAUTOLOGY in **NP**? Is it easy to verify that all assignments satisfy $f$?

# Example: TAUTOLOGY

TAUTOLOGY
**Instance**: A DNF-formula $f$
**Question:** Is it true that $f$ is a tautology, that is, is $f$ satisfied by all assignments?

An equivalent formulation of this problem:

TAUTOLOGY (VARIANT 2)
**Instance**: A DNF-formula $f$
**Question:** Is it true that there does not exist a truth assignment that un-satisfies $f$?

COMPLEMENT OF TAUTOLOGY
**Instance**: A DNF-formula $f$
**Question:** Does there exist a truth assignment that un-satisfies $f$?

Question: Have you seen the last problem with a different name?
Answer: SATISFIABILITY !!!

# **coNP**-completeness

## Definition

A language $\mathcal{L} \in$ **coNP** is said to be **coNP**-complete if, for any language $\mathcal{L}' \in$ **coNP**, we have $\mathcal{L}' \leq \mathcal{L}$.

## Theorem

*A language $\mathcal{L}$ is* **NP***-complete if and only if $\overline{\mathcal{L}}$ is* **coNP***-complete.*

Proof: We will show one direction, the other is similar. Let $\mathcal{L}$ be **NP**-complete and choose any $\mathcal{L}_0 \in$ **coNP**. Then $\overline{\mathcal{L}_0}$ is in **NP**. So there is a polynomial-time reduction $R$ from $\overline{\mathcal{L}_0}$ to $\mathcal{L}$ (i.e. $\overline{\mathcal{L}_0} \leq \mathcal{L}$). Then $R$ is also a reduction from $\mathcal{L}_0$ to $\overline{\mathcal{L}}$ :

$$x \in \mathcal{L}_0 \Leftrightarrow x \notin \overline{\mathcal{L}_0} \Leftrightarrow R(x) \notin \mathcal{L} \Leftrightarrow R(x) \in \overline{\mathcal{L}}.$$

□

## Corollary

*The problems* NO HAM CYCLE *and* TAUTOLOGY *are* **coNP***-complete.*

# NP vs. coNP

## Theorem

*If some* **coNP***-complete problem belongs to* **NP***, then* **NP** $=$ **coNP***.*

Proof: If $\mathcal{L} \in$ **NP** is **coNP**-complete then every $\mathcal{L}_0 \in$ **coNP** is polynomial-time reducible to $\mathcal{L}$ (i.e. $\mathcal{L}_0 \leq \mathcal{L}$). Thus, since $\mathcal{L} \in$ **NP**, it follows that $\mathcal{L}_0 \in$ **NP** and thus **coNP** $\subseteq$ **NP**.

By the theorem from the previous slide, $\overline{\mathcal{L}} \in$ **coNP** is **NP**-complete. Since every problem from **NP** is reducible to $\overline{\mathcal{L}}$, it follows that **NP** $\subseteq$ **coNP**. □

## Corollary

*If* **NP** $\neq$ **coNP** *then* TAUTOLOGY *and* NO HAM CYCLE *do not belong to* **NP***.*

# Remarks

- It holds that $\mathbf{P} = \mathbf{coP}$. (Why?)
- Clearly, if $\mathbf{P} = \mathbf{NP}$ then $\mathbf{NP} = \mathbf{coNP}$. However, it is still possible that $\mathbf{NP} = \mathbf{coNP}$ and $\mathbf{P} \neq \mathbf{NP}$.
- It is widely believed that $\mathbf{P} \neq \mathbf{NP}$ and $\mathbf{NP} \neq \mathbf{coNP}$.
- We also have $\mathbf{P} \subseteq \mathbf{NP} \cap \mathbf{coNP}$. It is not known whether this inclusion is strict, but most specialists tend to think it is.
- In many textbooks on complexity, the problem PRIMES (decide whether a given integer is prime) is given as an example of a problem in $\mathbf{NP} \cap \mathbf{coNP}$, and it was open for decades whether this problem is in $\mathbf{P}$.
- In 2002, M. Agrawal and two of his undergraduate students proved that PRIMES is in $\mathbf{P}$. Their algorithm is surprisingly simple, see `tinyurl.com/695dj7`.

# Around **NP**